

SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Modeling Object Relationships in a University Management System

CSA1107-OJECT ORIENTED ANALYSIS AND DESIGN

PRESENTED BY

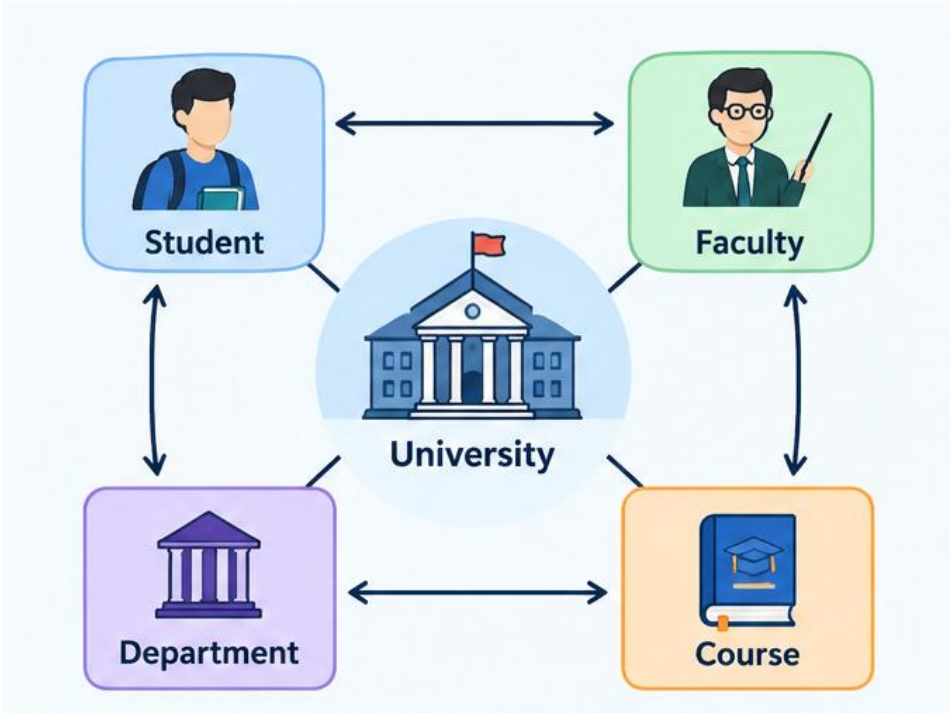
M. NISHANTH (192511154)

UNDER THE GUIDANCE OF

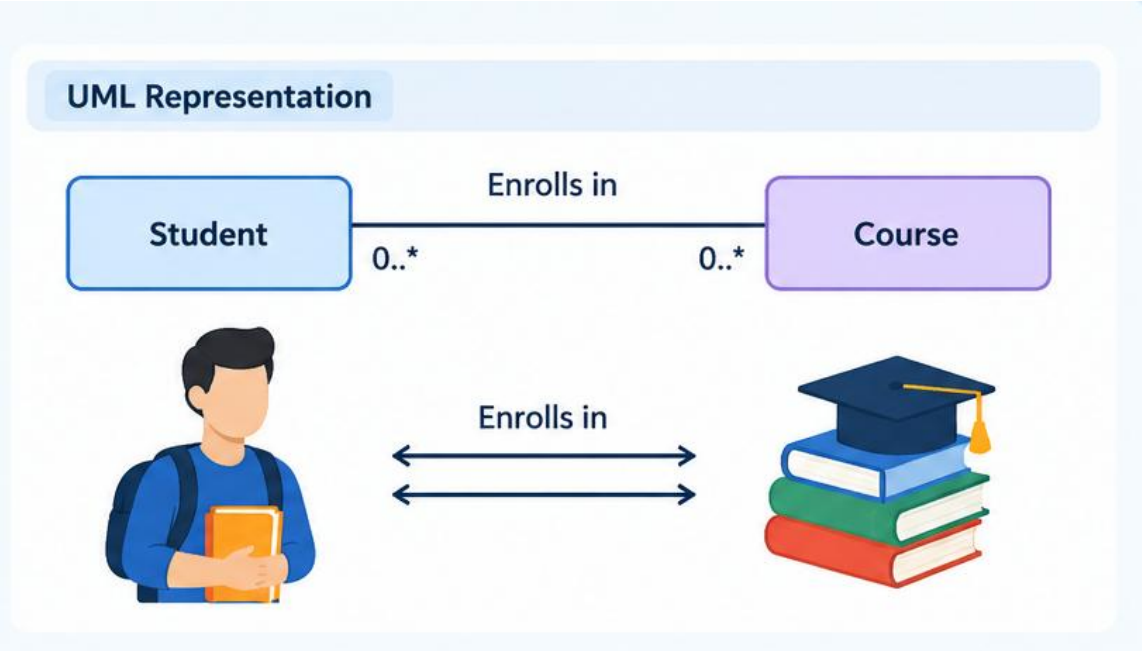
Dr. V. Madhan

INTRODUCTION

- Manages major entities such as **Students, Faculty, Departments, and Courses**.
- These entities interact with each other to perform university activities.
- **Object relationships** define how these entities are connected.
- UML provides different relationship types to represent these connections.
- The major relationships considered are **Association, Aggregation, and Composition**.
- Correct relationship modeling reduces unnecessary dependency between objects.



ASSOCIATION



Association Relationship

- Represents a **general interaction or connection** between two independent objects.
- Objects can exist independently of each other.
- Association can be **one-to-one, one-to-many, or many-to-many**.
- Example: **Student ↔ Course**
- A student can enroll in multiple courses.
- A course can contain multiple students.

UML Example

Student ——————————— Course

Enrolls in

Key Point:

Association = Objects interact with each other.

AGGREGATION AND COMPOSITION

- **Aggregation**
- Represents a **weak whole–part relationship**.
- The part can exist independently of the whole.
- Example: **Department ◇—— Faculty**
- If a department is removed, faculty members can still exist.
- UML symbol: **Hollow Diamond (◇)**
- Department ◇———————— Faculty

Composition

Represents a **strong whole–part relationship**.

The part's lifecycle depends on the whole.

Example: **Department ◆—— Course Offering**

If the department's course offerings are deleted, those offerings are also removed.

UML symbol: **Filled Diamond (◆)**

Department ◆———————— Course Offering

BENEFITS & CONCLUSION

- **How Proper Relationship Modeling Helps**
- **Flexibility**
- Reduces unnecessary dependency between objects.
- Makes it easier to modify system components.
- **Reusability**
- Independent objects can be reused in different parts of the system.
- Example: Faculty can work with different departments.
- **Maintainability**
- Clear relationships make the system easier to understand.
- Changes can be made without affecting unrelated objects.

Conclusion

- **Association** represents interaction.
- **Aggregation** represents weak whole–part relationships.
- **Composition** represents strong ownership.
- Choosing the appropriate relationship creates a **flexible, reusable, and maintainable** University Management System.

Thank You

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.



Any Questions ?